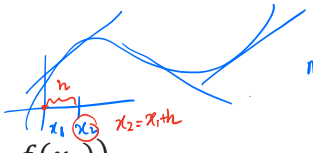
Derivatives

Derivatives are used to calculate velocity and acceleration, to estimate the rate of spread of a disease, to set levels of production so as to maximize efficiency, to find the best dimensions of a cylindrical can, and for many other applications.

In this chapter, we develop techniques to calculate derivatives easily and learn how to use derivatives to approximate complicated functions.

Tangent Lines and Derivative at a Point



$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

$= \frac{f(x_1 + h) - f(x_1)}{h}$

Finding a Tangent Line to the Graph of a Function

- Find a tangent line to an arbitrary curve $y = f(x)$ at a point $P(x_0, f(x_0))$.

We calculate the slope of the secant line through P and a nearby point $Q\ (x_0 + h, f(x_0 + h))$. Then we investigate the limit of the slope as $h \to 0$. If the limit exists, we call it the slope of the curve at P and define the tangent line at P to be the line through P having this slope.

Definition.

The **slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

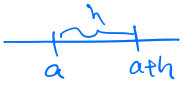
provided the limit exists.

The tangent line to the curve at P is the line throught P with this slope.

Example:

(a) Find the slope of the curve $y = \frac{1}{x}$ at any point $x = a \neq 0$. What is the slope at the point $x = -1$?

(b) Where does the slope equal $-\frac{1}{4}$?



(c) What happens to the tangent line to the curve at the point $\left(a, \frac{1}{a}\right)$ as a changes?

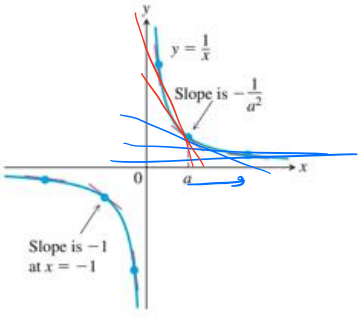


FIGURE 3.2 The tangent lines are steep when x is close to 0, and they become less steep as the point of tangency moves away (Example 1).

$$m = \frac{f(a+h) - f(a)}{h}$$

$$= \frac{\frac{1}{a+h} - \frac{1}{a}}{h}$$

$$= \frac{a - (a+h)}{h(a)(a+h)}$$

$$= \frac{-h}{ha(a+h)}$$

$$= \frac{-1}{a(a+h)}, \quad h \neq 0.$$

$$m \text{ at } x = a = \lim_{h \to 0} \frac{-1}{a(a+h)}$$

$$\text{slope at } x = a = \frac{-1}{a^2}.$$

$m|_{x=a} = -\frac{1}{a^2}$

$m|_{x=-1} = \frac{-1}{(-1)^2} = -1.$

(c) $m|_{x=a} = -\frac{1}{a^2}$

$(a, \frac{1}{a})$ changes

$a \to \infty$, $m \to 0$, tangent line = horizontal tangent line

$a \to 0$ $\quad m \to$ vertical tangent

$\frac{\Delta y}{\Delta t}$

Rates of Change: Derivative at a point

The expression

$$\frac{f(x_0+h)-f(x_0)}{h}, h \neq 0$$

is called the **difference quotient of f at x_0 with increment h.** If the difference quotient has a limit as h approaches zero, that limit is given a special name and notation.

Summary

We have been discussing slopes of curves, lines tangent to a curve, the rate of change of a function, and the derivative of a function at a point. All of these ideas are based on the same limit.

All of the following are interpretations for the limit of the difference quotient

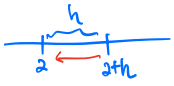
$$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}.$$

1. The slope of the graph of $y=f(x)$ at $x=x_0$
2. The slope of the tangent line to the curve $y=f(x)$ at $x=x_0$
3. The rate of change of $f(x)$ with respect to x at $x=x_0$
4. The derivative $f'(x_0)$ at $x=x_0$

Example:

Find an equation for the line tangent to the graph

$$g(x)=\frac{8}{x^2}, (2,2)$$



Slope at $(2,2)$ $= m|_{(2,2)} = \lim_{h\to 0}\frac{f(2+h)-f(2)}{(2+h)-2}$

$= \lim_{h\to 0}\frac{\frac{8}{(2+h)^2}-\frac{8}{2^2}}{h}$

$= \lim_{h\to 0}\frac{\frac{8}{(2+h)^2}-2}{h}$

$= \lim_{h\to 0}\frac{1}{h}\left(\frac{8-2(2+h)^2}{(2+h)^2}\right)$

$= \lim_{h\to 0}\frac{1}{h}\left(\frac{8-2(4+4h+h^2)}{(2+h)^2}\right)$

$= \lim_{h\to 0}\frac{1}{h}\left(\frac{-8h-2h^2}{(2+h)^2}\right)$

$= \lim_{h\to 0}\frac{1}{h}\left(\frac{-2h(4+h)}{(2+h)^2}\right)$

$= \lim_{h\to 0}\frac{-2(4+h)}{(2+h)^2}$

$= \frac{-8}{4} = -2.$

$y-2 = m(x-2)$

$y-2 = -2(x-2)$

$y = -2x+4+2$

$= -2x+6.$

$y+2x = 6.$

The Derivative of a Function

Definition. *Derivative Function*

The **derivative** of the function $f(x)$ with respect to the variable x is the function whose value at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

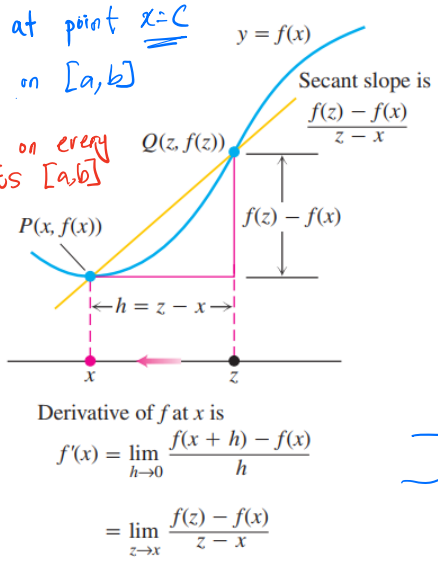
The domain of f' is the set of points in the domain of f for which the limit exists, and the domain f' may be the same or smaller than the domain of f.

If f' exists at a particular x, we say that f is **differentiable (has a derivative)** at x. If f' exists at every point in the domain of f, we call f **differentiable.**

f continuous at point x=c

f continuous on [a,b]

f cont on every points [a,b]



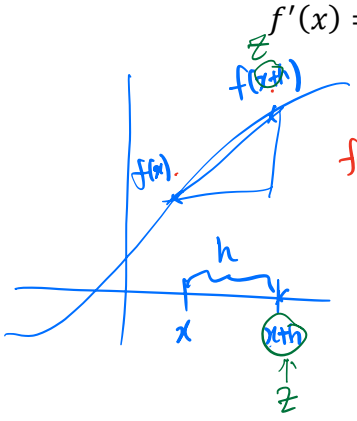
Derivative of f at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

Alternative Formula for the Derivative

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}.$$



$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{(x+h) - x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ ✓

$= \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$ ✓

$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

Calculating Derivatives from the Definition

The process of calculating a derivative is called **differentiation**. To emphasize the idea that differentiation is an operation performed on a function $y = f(x)$, we use notation

$$\frac{d}{dx} = d \div dx \quad \frac{d}{dx} f(x).$$

$\left(\frac{d}{dx}\right)$

$\frac{dy}{dx}$

$\frac{d}{dx}(y)$

As another way to denote the derivative $f'(x)$.

$\frac{dy}{dx} \neq dy \cdot \frac{1}{dx}$

$f'(x)$

Example. *Applying the Definition*

Differentiate $f(x) = \frac{x}{x-1}$.

Solution.

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

$$= \lim_{z \to x} \frac{\frac{z}{z-1} - \frac{x}{x-1}}{z - x}$$

$$= \lim_{z \to x} \left(\frac{1}{z-x}\right)\left(\frac{z(x-1) - x(z-1)}{(z-1)(x-1)}\right)$$

$$= \lim_{z \to x} \left(\frac{1}{z-x}\right)\left(\frac{zx - z - zx + x}{(z-1)(x-1)}\right)$$

$$f'(x) = \lim_{z \to x} \left(\frac{1}{z-x}\right)\left(\frac{-(z-x)}{(z-1)(x-1)}\right)$$

$$= \lim_{z \to x} \frac{-1}{(z-1)(x-1)}$$

$$= \frac{-1}{(x-1)^2}.$$

Example. Derivative of the Square Root Function

(a) Find the derivative of $y = \sqrt{x}$ for $x > 0$.

(b) Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

Solution.

(a) $y = \sqrt{x}$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h \to 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h}\right)\left(\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}\right)$$

$$= \lim_{h \to 0} \left(\frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})}\right)$$

$$= \lim_{h \to 0} \frac{h^1}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \left(\frac{1}{\sqrt{x+h} + \sqrt{x}}\right) = \frac{1}{2\sqrt{x}}.$$

(b) $\left.\frac{dy}{dx}\right|_{x=4} = \frac{1}{2\sqrt{4}}$

$$= \frac{1}{4}$$

$y = \sqrt{4} = 2$ where $x = 4$

$$y - 2 = \frac{1}{4}(x - 4)$$

$$4y - 8 = x - 4$$

$$4y = x + 4$$

Extra Example: $f(x) = e^x$

1. Suppose that f has the property $f(x + y) = f(x)f(y)$ for all values of x and y and that $f(0) = f'(0) = 1$. Show that f is differentiable and $f'(x) = f(x)$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x) \cdot f(h) - f(x)}{h}$$

$$= \lim_{h \to 0} f(x)\left(\frac{f(h) - 1}{h}\right)$$

$$= \lim_{h \to 0} f(x)\left(\frac{f(h) - 1}{h}\right)$$

$f(0) = f'(0) = 1$

$$f'(0) = \lim_{h \to 0} f(0)\left(\frac{f(h) - 1}{h}\right)$$

$$1 = \lim_{h \to 0} 1\left(\frac{f(h) - 1}{h}\right)$$

$$\Rightarrow \lim_{h \to 0}\left(\frac{f(h) - 1}{h}\right) = 1$$

$= f(x) \lim_{h\to 0}\left(\frac{f(h)-1}{h}\right)$

$= f(x)$. $\Rightarrow$ $f'(x) = f(x)$.

Notations

There are many ways to denote the derivative of a function $y = f(x)$, where the independent variable is x and the dependent variable is y.

Some common alternative notations for the derivative are

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = D(f)(x) = D_x f(x)$$

Note: The symbols d/dx and D indicate the operation of differentiation. We called **differentiation operators**.

** Interesting fact: The "prime" notations y' and f' come from notations that Newton used for derivatives. The d/dx notations are similar to those used by Leibniz.

To indicate the value of a derivative at a specified number $x = a$, we use the notation

$$f'(a) = \frac{dy}{dx}\Big|_{x=a} = \frac{df}{dx}\Big|_{x=a} = \frac{d}{dx}f(x)\Big|_{x=a}$$

For example,

$$f'(4) = \frac{d}{dx}\sqrt{x}\Big|_{x=4} = \frac{1}{2\sqrt{x}}\Big|_{x=4} = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

Differentiability on an Interval; One-Sided Derivatives

A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval.

It is **differentiable on a closed interval** $[a, b]$, if it is differentiable on the interior(a, b), and if the limits

$\lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}$ Right-hand derivative at a

$\lim_{h\to 0^-} \frac{f(b+h)-f(b)}{h}$ Left-hand derivative at a

A function has a derivative at an interior point if and only if it has left-hand and right-hand derivatives there, and these one-sided derivatives are equal.

Example. $y = |x|$ *Is Not Differentiable at the Origin.*

Show that the function $y = |x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$ but has no derivative at $x = 0$.

Solution.

$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

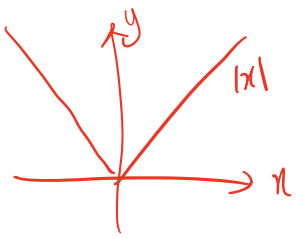
$\frac{d}{dx}|x|\Big|_{x=0} = f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{f(h)-f(0)}{h}$

$f'(0) = \lim_{h\to 0} \frac{|h|-|0|}{h} = \lim_{h\to 0} \frac{|h|}{h}$

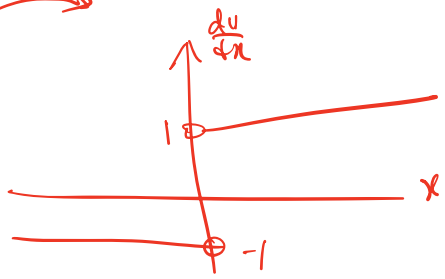
$\lim_{h\to 0^+} \frac{|h|}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1$

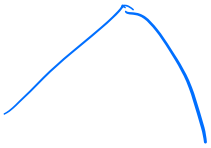
$\lim_{h\to 0^-} \frac{|h|}{h} = \lim_{h\to 0^-} \frac{-h}{h} = -1$

$\Rightarrow \lim_{h\to 0} \frac{|h|}{h}$ doesn't exist $\Rightarrow$ f is not differentiable at $x=0$.



$\frac{d}{dx}|x| = \begin{cases} 1, & x>0 \\ -1, & x<0 \end{cases}$





Extra Example:

1. Show that $f(x) = |5x - 3|$ is not differentiable at $x = \frac{3}{5}$.

2. Let $f(x) = \begin{cases} 1 + 2x, & x < 1; \\ 4 - x^2, & x \geq 1. \end{cases}$ Is f differentiable at 1? Explain.

(1) $f(x) = |5x-3| = \begin{cases} 5x-3, & x \geq \frac{3}{5} \\ -(5x-3), & x < \frac{3}{5} \end{cases}$

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$f'(\frac{3}{5}) = \lim_{h\to 0} \frac{|5(\frac{3}{5}+h)-3| - |5(\frac{3}{5})-3|}{h}$

$f'(\frac{3}{5}) = \lim_{h\to 0} \frac{5|h|-0}{h}$

$= \lim_{h\to 0} 5\frac{|h|}{h}$

$= 5 \lim_{h\to 0} \frac{|h|}{h}$

$\lim_{h\to 0^+} \frac{|h|}{h} = 1$ $\quad \lim_{h\to 0^-} \frac{|h|}{h} = -1$

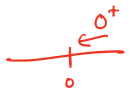
$\lim_{h\to 0} \frac{|h|}{h}$ doesn't exist $\Rightarrow$ f is not diff. at $x=\frac{3}{5}$.

(2) $f(x) = \begin{cases} 1+2x, & x<1 \\ 4-x^2, & x \geq 1 \end{cases}$

$f'(1) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$

$= \lim_{h\to 0} \frac{f(1+h)-(4-1^2)}{h}$

$= \lim_{h\to 0} \frac{f(1+h)-3}{h}$



$\lim_{h\to 0^+} \frac{f(1+h)-3}{h}$

$= \lim_{h\to 0^+} \frac{4-(1+h)^2-3}{h}$

$= \lim_{h\to 0^+} \frac{1-(1+2h+h^2)}{h}$

$= \lim_{h\to 0^+} \frac{-h(2+h)}{h} = -2$

$\lim_{h\to 0^-} \frac{f(1+h)-3}{h}$

$= \lim_{h\to 0^-} \frac{1+2(1+h)-3}{h}$

$= \lim_{h\to 0^-} \frac{2h}{h}$

$= 2$.

$\Rightarrow \lim_{h\to 0} \frac{f(1+h)-3}{h}$ doesn't exist

$\Rightarrow f$ not diff at $x=1$

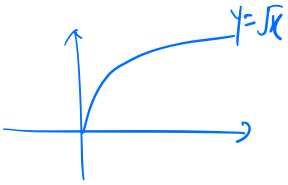
Example. $y = \sqrt{x}$ *Is Not Differentiable at* $x = 0$

$f'(0) = \lim_{h\to 0} \frac{\sqrt{0+h}-\sqrt{0}}{h}$

$= \lim_{h\to 0} \frac{\sqrt{h}}{h}$

$\lim_{h\to 0^+} \frac{\sqrt{h}}{h} = \lim_{h\to 0^+} \frac{1}{\sqrt{h}} = \infty$

$\downarrow$ vertical tangent.

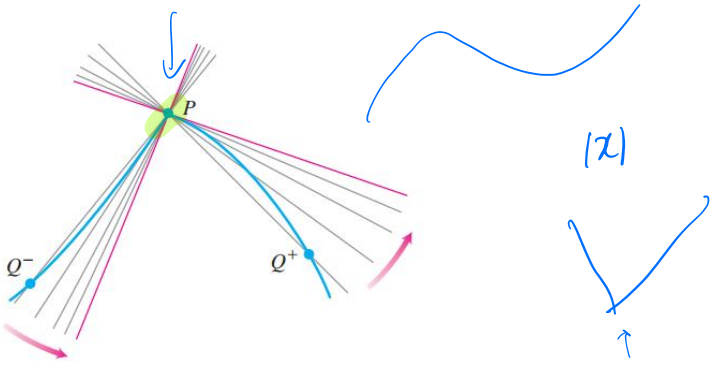


Extra Note: Since the slopes of the secant lines joining the origin to the points $(h, \sqrt{h})$ on a graph of $y = \sqrt{x}$ approach ∞, the graph has a *vertical tangent* at the origin.

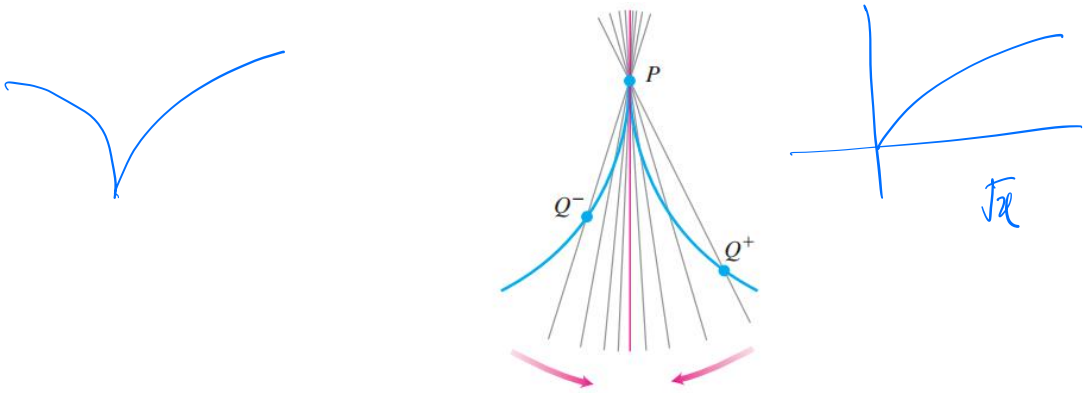
When Does a Function *NOT* have a Derivative at a Point?

Differentiability is a "smoothness" condition on the graph of f. A function whose graph is otherwise smooth will fail to have a derivative at a point for several reasons, such as at points where the graph has

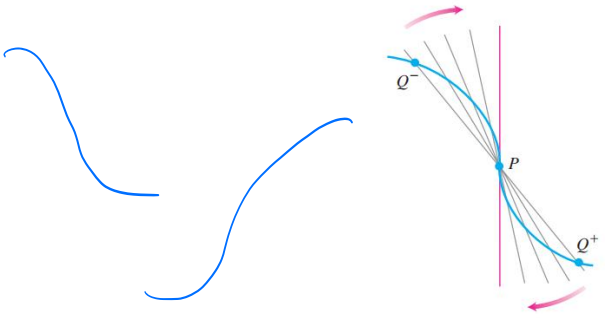
(i) a ***corner***, where the one-sided derivatives differ.



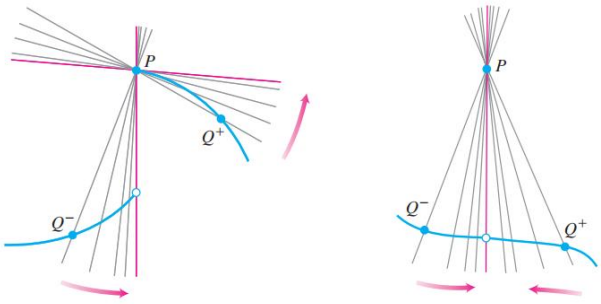
(ii) a ***cusp***, where the slope of PQ approaches ∞ from one side and $-\infty$ from the other.



(iii) a ***vertical tangent***, where the slope of PQ approaches ∞ from both sides or approaches $-\infty$ from both sides (here, $-\infty$).



(iv) a ***discontinuity***.



Differentiable Functions Are Continuous

A function is continuous at every point where it has a derivative.

if p, then q

$p \Rightarrow q \equiv \sim q \Rightarrow \sim p$.

f diff at $x=c \Rightarrow$ f cont. $x=c$

$\Rightarrow$ f not cont at $x=c \Rightarrow$ f not diff at $x=c$.

Theorem. *Differentiability Implies Continuity*

If f has a derivative at $x = c$, then f is continuous at $x = c$.

Proof. Given that $f'(x)$ exists, we must show that $\lim_{x\to c} f(x) = f(c)$, or equivalently, that $\lim_{h\to 0} f(c+h) = f(c)$. If $h \neq 0$, then

$|x|$ at $x=0$.

$\frac{d}{dx}|x| = \frac{d}{dx}\sqrt{x^2}$

$=$ ____

$$f(c+h) = f(c) + \big(f(c+h) - f(c)\big)$$
$$= f(c) + \frac{\big(f(c+h) - f(c)\big)}{h} \cdot h$$

Now take limits as $h \to 0$.

$$\lim_{h\to 0} f(c+h) = \lim_{h\to 0} f(c) + \lim_{h\to 0} \frac{\big(f(c+h) - f(c)\big)}{h} \cdot \lim_{h\to 0} h$$
$$= f(c) + f'(c) \cdot 0$$
$$= f(c) + 0$$
$$= f(c).$$

Caution: *The converse of Theorem is false*. A function need not have a derivative at a point where it is continuous.

For example. $y = |x|$ is a continuous function but it is not differentiable at $x = 0$.

Differentiation Rules

This section introduces a few rules that allow us to differentiate a great variety of functions. By proving these rules here, we can differentiate functions without having to apply the definition of the derivative each time.

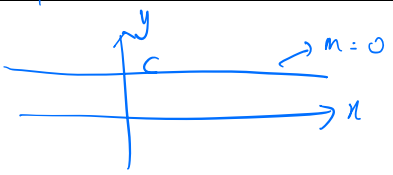
Powers, Multiples, Sums, and Differences

$y = c$

Rule 1. *Derivative of a Constant Function*

If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$



Example.

$$\frac{d}{dx}(8) = 0, \qquad \frac{d}{dx}\left(-\frac{\pi}{2}\right) = 0, \qquad \frac{d}{dx}(\sqrt{3}) = 0$$

Rule 2. *Power Rule for Positive Integers*

If n has a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Proof. The formula

$$z^n - x^n = (z - x)(z^{n-1} + z^{n-2}x + \cdots + zx^{n-2} + x^{n-1})$$

By using the alternative form for the definition of the derivative,

$$\begin{aligned} f'(x) &= \lim_{z \to x} \frac{f(z) - f(x)}{z - x} = \lim_{z \to x} \frac{z^n - x^n}{z - x} \\ &= \lim_{z \to x} (z^{n-1} + z^{n-2}x + \cdots + zx^{n-2} + x^{n-1}) \\ &= nx^{n-1} \end{aligned}$$

Example.

f	x	x^2	x^3	x^4	x^5	…
f'	1	$2x$	$3x^2$	$4x^3$	$5x^4$	…

Rule 3. *Constant Multiple Rules*

If u is a differentiable function of x, and c is a constant, then

$$\frac{d}{dx}(cu) = c\frac{du}{dx}.$$

Particularly, if n is a positive integer, then

$$\frac{d}{dx}(cx^n) = cnx^{n-1}.$$

Example.

(a) $\frac{d}{dx}(3x^2) = 3(2x) = 6x$

(b) **A useful special case**

The derivative of the negative of a differentiable function u is the negative of the function's derivative. Apply Rule 3 with $c = -1$ gives

$$\frac{d}{dx}(-u) = -1 \cdot \frac{d}{dx}(u) = -\frac{du}{dx}.$$

In particular, if y is a differentiable function, then its derivative

$$\frac{d}{dx}(y) = \frac{dy}{dx}.$$

$\frac{d}{dx}(u_1+u_2+\dots+u_n) = \frac{du_1}{dx}+\frac{du_2}{dx}+\dots+\frac{du_n}{dx}$

$\frac{d}{dx}(u+v+w) = \frac{du}{dx}+\frac{dv}{dx}+\frac{dw}{dx}$

Rule 4. *Derivative Sum Rules*

If u and v are differentiable function of x, then their sum $u + v$ is differentiable at every point where u and v are both differentiable. At such points,

$$\frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx}.$$

$u = f(x)$, $v = g(x)$

$\frac{d}{dx}(u+v) = \frac{du}{dx}+\frac{dv}{dx}$

Proof. We let $f(x) = u(x) + v(x)$.

$$\begin{aligned}\frac{d}{dx}\big(u(x)+v(x)\big) &= \lim_{h\to 0}\frac{\big(u(x+h)+v(x+h)\big)-\big(u(x)+v(x)\big)}{h}\\ &= \lim_{h\to 0}\frac{\big(u(x+h)-u(x)\big)-\big(v(x+h)-v(x)\big)}{h}\\ &= \lim_{h\to 0}\frac{\big(u(x+h)-u(x)\big)}{h} - \lim_{h\to 0}\frac{\big(v(x+h)-v(x)\big)}{h}\\ &= \frac{du}{dx}+\frac{dv}{dx}.\end{aligned}$$

Example. *Derivative of a Sum*

$y = x^4 + 12x$

$\frac{dy}{dx} = 4x^3 + 12$

$\frac{d}{dx}(x^4+12x) = \frac{d}{dx}(x^4) + \frac{d}{dx}(12x)$

$= \frac{d}{dx}(x^4) + 12\frac{d}{dx}(x)$

$= 4x^3 + 12(1)$.

Note. We can combine Rule 3 and Rule 4, it gives the Difference Rule.

$$\frac{d}{dx}(u-v) = \frac{du}{dx} - \frac{dv}{dx}$$

$\frac{d}{dx}(u-v)$ $c=-1$

$= \frac{d}{dx}(u+(-v))$

$= \frac{du}{dx} + \frac{d}{dx}(-v)$

$= \frac{du}{dx} - \frac{dv}{dx}$

The Sum Rule (Rule 4) also extends to sums of more than two functions, as long as there are only finitely many functions in the sum.

Let $u_1, u_2, \dots, u_n$ are differentiable at x, then so is $u_1 + u_2 + \cdots + u_n$, and

$$\frac{d}{dx}(u_1+u_2+\cdots+u_n) = \frac{du_1}{dx}+\frac{du_2}{dx}+\cdots+\frac{du_n}{dx}.$$

$y = x^3 + \frac{4}{3}x^2 - 5x + 1$

$\frac{dy}{dx} = 3x^2 + \frac{4}{3}(2x) - 5(1) + 0$

$= 3x^2 + \frac{8}{3}x - 5$.

Example. *Derivative of a Polynomial*

$$y = x^3 + \frac{4}{3}x^2 - 5x + 1$$
$$\frac{dy}{dx} = 3x^2 + \frac{8}{3}x - 5$$

Note. All polynomials are differentiable everywhere.

Example. Finding Horizontal Tangents

$m = 0 \left(\frac{dy}{dx} = 0\right)$.

Does the curve $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so, where?

Solution.

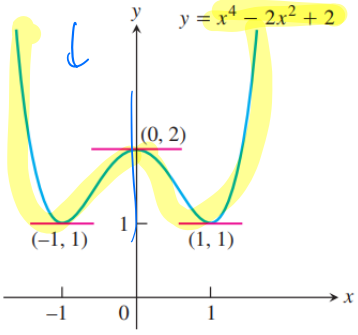
$y = x^4 - 2x^2 + 2$

$\frac{dy}{dx} = 4x^3 - 4x$

$\frac{dy}{dx} = 0 \Rightarrow \quad 4x^3 - 4x = 0$

$4x(x^2 - 1) = 0$

$x = 0, \quad x = 1, \quad x = -1$



$u = f(x)$
$v = g(x)$
$w = h(x)$

$\frac{d}{dx}(uvw) = uv\frac{dw}{dx} + uw\frac{dv}{dx} + vw\frac{du}{dx}$

$\frac{d}{dx}(uvwz) = vwz\frac{du}{dx} + uwz\frac{dv}{dx} + uvz\frac{dw}{dx} + uvw\frac{dz}{dx}$

Products and Quotients

Rule 5. *Derivative Product Rules*

If u and v are differentiable function of x, then their product uv and

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

In function notation,

$$\frac{d}{dx}(f(x)g(x)) = f(x)g'(x) + g(x)f'(x).$$

Proof. We let $f(x) = u(x) + v(x)$.

$$\frac{d}{dx}(uv) = \lim_{h\to 0}\frac{\big(u(x+h)v(x+h)\big) - \big(u(x)v(x)\big)}{h}$$

$$= \lim_{h\to 0}\frac{u(x+h)v(x+h) - \boldsymbol{u(x+h)v(x) + u(x+h)v(x)} - u(x)v(x)}{h}$$

$$= \lim_{h\to 0} u(x+h)\cdot\left(\frac{v(x+h)-v(x)}{h}\right) + v(x)\cdot\lim_{h\to 0}\frac{u(x+h)-u(x)}{h}$$

As $h \to 0, u(x+h) \to u(x)$,

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

Example. *Using Product Rule*

Find derivative of $y = \frac{1}{x}\left(x^2 + \frac{1}{x}\right)$.

$y = \frac{1}{x}\left(x^2+\frac{1}{x}\right) = x + \frac{1}{x^2}$

$\frac{dy}{dx} = 1 - \frac{2}{x^3}$

Solution.

$y = \left(\frac{1}{x}\right)\left(x^2+\frac{1}{x}\right)$

$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$ $= \left(x^2+\frac{1}{x}\right)\left(\frac{-1}{x^2}\right) + \left(\frac{1}{x}\right)\left(2x - \frac{1}{x^2}\right)$

$= -1 + \frac{-1}{x^3} + 2 - \frac{1}{x^3}$

$= 1 - \frac{2}{x^3}$

Example.

Find the derivative of $y = (x^2+1)(x^3+3)$.

$y = x^5 + 3x^2 + x^3 + 3$

$\frac{dy}{dx} = 5x^4 + 6x + 3x^2$

$= 5x^4 + 3x^2 + 6x$

Solution.

$y = (x^2+1)(x^3+3)$

$\frac{dy}{dx} = 2x(x^3+3) + (x^2+1)(3x^2)$

$= 2x^4 + 6x + 3x^4 + 3x^2 = 5x^4 + 3x^2 + 6x$

Rule 6. *Derivative Quotient Rules*

If u and v are differentiable function of x and if $v(x) \neq 0$, then the quotient u/v is differentiable at x, and

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

In function notation,

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}.$$

Proof. We let $f(x) = u(x) + v(x)$.

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \lim_{h\to 0} \frac{\left(\frac{u(x+h)}{v(x+h)}\right) - \left(\frac{u(x)}{v(x)}\right)}{h}$$

$$= \lim_{h\to 0} \frac{v(x)u(x+h) - u(x)v(x+h)}{hv(x+h)v(x)}$$

$$= \lim_{h\to 0} \frac{v(x)u(x+h) - \boldsymbol{v(x)u(x) + v(x)u(x)} - u(x)v(x+h)}{hv(x+h)v(x)}$$

$$= \lim_{h\to 0} \frac{v(x)[u(x+h) - u(x)] + u(x)[v(x) - v(x+h)]}{hv(x+h)v(x)}$$

$$= \lim_{h\to 0} \frac{v(x)\dfrac{u(x+h)-u(x)}{h} + u(x)\dfrac{v(x)-v(x+h)}{h}}{v(x+h)v(x)}$$

As $h \to 0, v(x+h) \to v(x)$,

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}.$$

Example. *Use the Quotient Rule*

Find the derivative of

$$y = \frac{t^2 - 1}{t^2 + 1}.$$

Solution.

$$\frac{dy}{dt} = \frac{(t^2+1)(2t) - (t^2-1)(2t)}{(t^2+1)^2}$$

$$= \frac{4t}{(t^2+1)^2}$$

$y = f(t)$, u, v

$$y = \frac{t^2-1}{t^2+1} = \frac{t^2+1-2}{t^2+1} = 1 - \frac{2}{t^2+1}$$

$$\frac{dy}{dt} = \frac{-2(-1)}{(t^2+1)^2}(2t) = \frac{4t}{(t^2+1)^2}$$

Negative Integer Powers of x

The Power Rule for negative integers is the same as the rule for positive integers.

***Rule** 7. Power Rule for Negative Integers*

If n has a *negative* integer and $x \neq 0$, then

n = negative integer

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Example.

(a) $\dfrac{d}{dx}\left(\dfrac{1}{x}\right) = \dfrac{d}{dx}(x^{-1}) = -1(x^{-1-1}) = -x^{-2} = -\dfrac{1}{x^2}.$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}\left(x^{-1}\right) = -1\left(x^{-2}\right) = \frac{-1}{x^2}$$

$\frac{d}{dx}\left(\frac{4}{x^3}\right) = \frac{d}{dx}(4x^{-3}) = 4\frac{d}{dx}(x^{-3}) = 4(-3x^{-4})$

$= \frac{-12}{x^4}$

(b) $\dfrac{d}{dx}\left(\dfrac{4}{x^3}\right) = 4\dfrac{d}{dx}(x^{-3}) = 4(-3)(x^{-3-1}) = -12x^{-4} = -\dfrac{12}{x^4}.$

Example. *Tangent to a Curve*

Find an equation for the tangent to the curve

$$y = x + \frac{2}{x}$$

at the point $(1,3)$.

Solution.

$y = x + \frac{2}{x}$

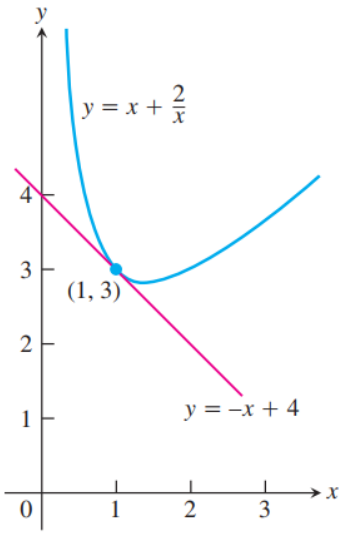
$\frac{dy}{dx} = 1 - \frac{2}{x^2}$

$\left.\frac{dy}{dx}\right|_{x=1} = 1 - \frac{2}{1^2} = -1$

$y - 3 = -1(x-1)$

$y = -x + 1 + 3$

$y = 4 - x \quad (y + x = 4).$



Note. The choice of which rules to use in solving a differentiation problem can make a difference in how much work you have to do.

Example. *Choosing Which Rule to Use.*

Find the derivative without using Quotient Rule.

$$y = \frac{(x-1)(x^2-2x)}{x^4}$$

$y = \frac{(x-1)(x^2-2x)}{x^4}$

$= \frac{x^3 - 2x^2 - x^2 + 2x}{x^4}$

$= \frac{x^3 - 3x^2 + 2x}{x^4}$

$y = \frac{1}{x} - \frac{3}{x^2} + \frac{2}{x^3}$

$\frac{dy}{dx} = \frac{-1}{x^2} + \frac{6}{x^3} - \frac{6}{x^4}$

Solution. We expand the numerator and divide by x^4,

$$y = \frac{(x-1)(x^2-2x)}{x^4} = \frac{x^3 - 3x^2 + 2x}{x^4} = x^{-1} - 3x^{-2} + 2x^{-3}.$$

Then use the Sum and Power Rules:

$$\frac{dy}{dx} = -x^{-2} + 6x^{-3} - 6x^{-4} = \frac{-1}{x^2} + \frac{6}{x^3} - \frac{6}{x^4}.$$

$\sin(\sin x)$

$\sin^2 x = (\sin x)^2$

y, $\frac{dy}{dx}$, $\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d^2y}{dx^2}$, $\frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)=\frac{d^3y}{dx^3}$, $\frac{d^ny}{dx^n}$

y'

$f'(x)$, $f''(x)$, $f'''(x)$, $f^{IV}(x)$, $f^{V}(x)$, $f^{VI}(x)$, $\cdots$

$f'(x)$, $f^{(2)}(x)$, $f^{(3)}(x)$, $f^{(4)}(x)\ldots$, $f^{(n)}(x)$

$f^2(x)$

$=(f\circ f)(x)$

Second- and Higher-Order Derivatives

We note that if $f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function. Similarly, if f' is differentiable, then we can differentiate f' to get a new function of x, denoted as f'', where $f''=(f')'$.

The function f'' is called the **second derivative** of f.

$D^n(f(x))$

Notation:

$$f''(x)=\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{dy'}{dx}=y''=D^2(f)(x)=D_x^2f(x).$$

Note: The symbol D^2 means the Operation of Differentiation is performed twice.

$y=x^6$

$y'=6x^5$

$y''=30x^4$

$y'''=120x^3$

$y^{(4)}=360x^2$

$y^{(5)}=720x$

$y^{(6)}=720$

$y^{(7)}=0.$

$\vdots$

$y^{(n)}=0.$

Example. If $y=x^6$, then $y'=6x^5$and we have

$$y''=\frac{d}{dx}(6x^5)=30x^4.$$

If y'' is differentiable, its derivative, $y'''=\frac{dy''}{dx}=\frac{d^3y}{dx^3}$ is the third derivative of y with respect to x. The names continue as you imagine, with

$$y^{(n)}=\frac{d}{dx}y^{(n-1)}=\frac{d^ny}{dx^n}=D^ny$$

denoting the ***n*th derivative** of y with respect to x for any positive integer n.

$y=x^3-3x^2+2$

$\frac{dy}{dx}=3x^2-6x$

$\frac{d^2y}{dx^2}=6x-6$

$\frac{d^3y}{dx^3}=6$

$\frac{d^4y}{dx^4}=0$ $\cdots$ $\frac{d^ny}{dx^n}=0.$

Example. Finding Higher Derivatives

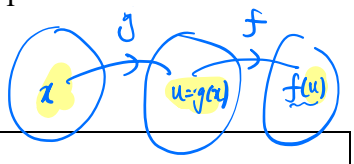
The first four derivatives of $y=x^3-3x^2+2$ are

$$y'=3x^2-6x$$
$$y''=6x-6$$
$$y'''=6$$
$$y^{(4)}=0.$$

The function has derivatives of all orders, the fifth and later derivatives all being zero.

The Chain Rule and Implicit Differentiation

The Chain Rule, which says that the derivative of the composite of two differentiable functions is the product of their derivatives evaluated at appropriate points. The Chain Rule is one of the most important and widely used rules of differentiation.



Theorem. *The Chain Rule.*

If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x, and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

g diff. at x
u=g(x)
y=f(u)
f diff u=g(x)

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.

where dy/du is evaluated at $u = g(x)$.

Example. *Applying the Chain Rule.*

An object moves along the x-axis so that its positive at any time $t \geq 0$ is given by $x(t) = \cos(t^2 + 1)$. Find the velocity of the object as a function of t.

Solution.

$v = \frac{d}{dt} x$

$\frac{d}{dx}(f(g(x)))$

$= f'(g(x)) \cdot g'(x)$

$x(t) = \cos(t^2+1)$

$x'(t) = -\sin(t^2+1) \cdot (2t)$

$= -2t \sin(t^2+1)$.

$0 \longrightarrow$

$x(t) = \cos(t^2+1)$

$u = t^2+1 \rightarrow u = f(t)$

$x = \cos u \quad x = g(u)$

$\frac{dx}{dt} = \frac{dx}{du} \cdot \frac{du}{dt}$

$= -\sin u \cdot (2t) = -2t \sin(t^2+1)$.

"Outside-Inside" Rule

It is helpful to think about Chain Rule this way:

If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Note: In words, differentiate the "outside" function f and evaluate it at the "inside" function $g(x)$ left alone; then multiple by the derivative of the "inside" function.

$\frac{d}{dx}\left(\sin(x^2+x)\right) = \cos(x^2+x)\cdot(2x+1)$
$= (2x+1)\cos(x^2+x)$

Example. *Differentiating from the Outside In*

Differentiate $\sin(x^2+x)$ with respect to x.

Solution.

$$\frac{d}{dx}\sin(x^2+x) = \cos(x^2+x)\cdot(2x+1)$$

Inside — Inside left it alone — derivative of the inside

Example.

Find the derivative of $g(t) = \tan(5-\sin 2t)$.

$g(x) = \tan(5-\sin 2t)$ (outside / inside)
$g'(x) = \sec^2(5-\sin 2t)\cdot[-\cos(2t)\cdot 2]$
$= -2\cos(2t)\sec^2(5-\sin 2t)$

Solution.

$$g'(t) = \sec^2(5-2\sin 2t)\cdot(0-\cos(2t)\cdot 2)$$
$$= -2(\cos 2t)(\sec^2(5-2\sin 2t))$$

$\frac{d}{dx}\left((5x^3-x^4)^7\right) = 7(5x^3-x^4)^6\cdot(15x^2-4x^3)$.

$\frac{d}{dx}\left(\frac{1}{3x-2}\right) = \frac{d}{dx}\left((3x-2)^{-1}\right)$
$= -1(3x-2)^{-2}\cdot(3)$
$= \frac{-3}{(3x-2)^2}$

$\frac{d}{dx}\left(\sqrt{1+\sqrt{x}}\right) = \frac{1}{2\sqrt{(1+\sqrt{x})}}\cdot\left(\frac{1}{2\sqrt{x}}\right)$

Example.

(a) $\frac{d}{dx}(5x^3-x^4)^7 = 7(5x^3-x^4)^6(15x^2-4x^3)$

(b) $\frac{d}{dx}\left(\frac{1}{3x-2}\right) = \frac{d}{dx}((3x-2)^{-1}) = -1(3x-2)^{-2}(3) = -\frac{3}{(3x-2)^2}$

Example. *Finding Tangent Slopes*

(a) Find the slope of the line tangent to the curve $y = \sin^5 x$ at the point where $x = \frac{\pi}{3}$.

(b) Show that the slope of every line tangent to the curve $y = 1/(1-2x)^3$ is positive.

$\frac{d}{dx}\left(\sqrt{x+\sqrt{x+\sqrt{x}}}\right)$
$= \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\cdot\left(1+\frac{d}{dx}\sqrt{x+\sqrt{x}}\right)$
$= \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\cdot\left(1+\frac{1}{2\sqrt{x+\sqrt{x}}}\cdot\left(1+\frac{d}{dx}\sqrt{x}\right)\right)$
$= \left(\frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\right)\cdot\left(1+\frac{1}{2\sqrt{x+\sqrt{x}}}\cdot\left(1+\frac{1}{2\sqrt{x}}\right)\right)$.

Solution.

(a) $y = \sin^5 x = (\sin x)^5$

$\frac{dy}{dx} = 5\sin^4 x\cdot\cos x$

$\left.\frac{dy}{dx}\right|_{x=\frac{\pi}{3}} = 5\left(\sin\left(\frac{\pi}{3}\right)\right)^4\cos\left(\frac{\pi}{3}\right)$
$= 5\left(\frac{\sqrt{3}}{2}\right)^4\left(\frac{1}{2}\right)$
$= \frac{5(9)}{32} =$ ___.

(b) $y = \frac{1}{(1-2x)^3}$

$\frac{dy}{dx} = \frac{-3}{(1-2x)^4}(-2)$
$= \frac{6}{(1-2x)^4} > 0$

f' ⟵ $y = f(x)$, $\frac{dy}{dx} =$ ___ ⟶ $\int f\,dx$

$y^2 - x = 0$
$2y\frac{dy}{dx} - 1 = 0$
$2y\frac{dy}{dx} = 1$
$\frac{dy}{dx} = \frac{1}{2y}$

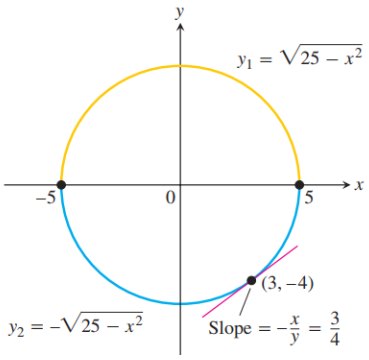
$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$.

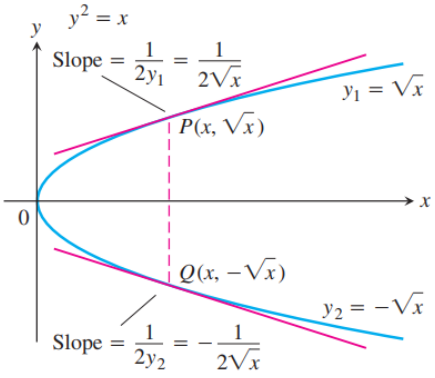
$y = f(x)$

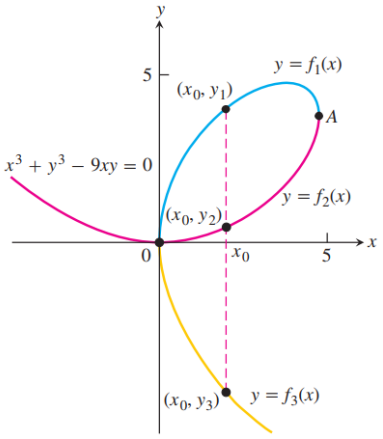
Implicit Differentiation

when we encounter equations like

$$x^2 + y^2 - 25 = 0, \qquad y^2 - x = 0, \qquad x^3 + y^3 - 9xy = 0.$$







These equations define an *implicit* relation between the variables x and y. In some cases, we may be able to solve such an equation for y as an explicit function (or even several functions) of x.

When we cannot put an equation $F(x, y) = 0$ in the form $y = f(x)$ to differentiate it in the usual way, we may still be able to find dy/dx by *implicit differentiation*.

This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for y'.

Implicitly Defined Functions

Example. *Differentiating Implicitly*

Find dy/dx if $y^2 = x$.

Solution. The equation $y^2 = x$ defines two differentiable functions of x that we can actually find, namely $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$. This follows that, $\frac{dy_1}{dx} = \frac{1}{2\sqrt{x}}$ and $\frac{dy_2}{dx} = -\frac{1}{2\sqrt{x}}$.

Besides that, we can also simply differentiate both sides of the equation $y^2 = x$ with respect to x, treating $y = f(x)$ as a differentiable function of x:

$$y^2 = x \;\Rightarrow\; 2y\frac{dy}{dx} = 1 \;\Rightarrow \frac{dy}{dx} = \frac{1}{2y}.$$

This one formula gives the derivatives we calculated for *both* explicit solutions $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$:

$$\frac{dy_1}{dx} = \frac{1}{2y_1} = \frac{1}{2\sqrt{x}} \qquad \text{and} \qquad \frac{dy_2}{dx} = \frac{1}{2y_2} = \frac{1}{2(-\sqrt{x})} = -\frac{1}{2\sqrt{x}}.$$

LHS $= 3^2 + (-4)^2 = 9 + 16 = 25 =$ RHS

Example. *Slope of a Circle at a Point*

Find the slope of circle $x^2 + y^2 = 25$ at the point $(3, -4)$.

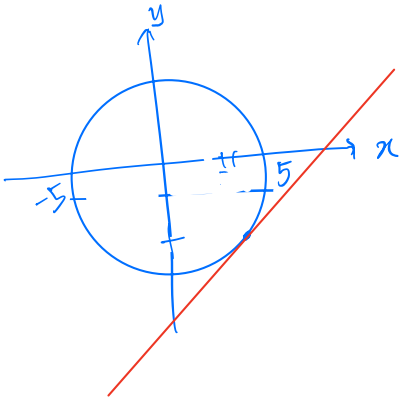
Solution.

$x^2 + y^2 = 25$

$2x + 2y \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{-x}{y}$

$\left.\frac{dy}{dx}\right|_{(3,-4)} = \frac{-3}{-4} = \frac{3}{4}.$



Example. *Differentiating Implicitly*

Find dy/dx if $y^2 = x^2 + \sin(xy)$.

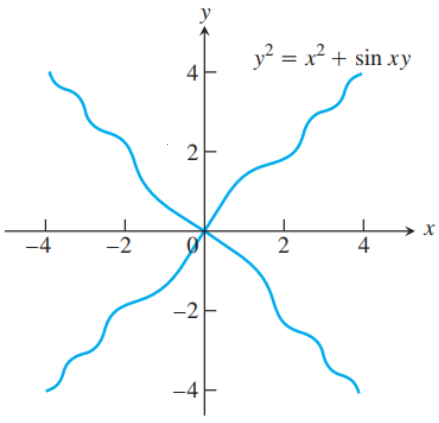
Solution.

$y^2 = x^2 + \sin(xy)$

$2y \frac{dy}{dx} = 2x + \cos(x \cdot y)\left[x \frac{dy}{dx} + y\right]$

$2y \frac{dy}{dx} - x\cos(xy) \frac{dy}{dx} = y\cos(xy) + 2x$

$\frac{dy}{dx} = \frac{y\cos(xy) + 2x}{2y - x\cos(xy)}.$



Implicit Differentiation

1. Differentiate both sides of the equation with respect to x, treating y as a differentiable function of x.

2. Collect the terms with dy/dx on one side of the equation.

3. Solve for dy/dx.

Example. *Tangent and Normal to the Folium of Descartes*

Show that the point (2,4) lies on the curve $x^3 + y^3 - 9xy = 0$. Then find the tangent and normal to the curve there.

Solution.

$LHS = 2^3 + 4^3 - 9(2)(4)$

$= 8 + 64 - 72$

$= 0$

$= RHS.$

$\Rightarrow$ (2,4) lies on curve $x^3 + y^3 - 9xy = 0$.

$x^3 + y^3 - 9xy = 0$

$3x^2 + 3y^2 \frac{dy}{dx} - 9\left(x\frac{dy}{dx} + y\right) = 0$

$\frac{dy}{dx} = \frac{9y - 3x^2}{3y^2 - 9x}$

$= \frac{3y - x^2}{y^2 - 3x}$

Slope tangent $= \left.\frac{dy}{dx}\right|_{(2,4)} = \frac{3(4) - 2^2}{4^2 - 3(2)}$

$= \frac{12 - 4}{16 - 6}$

$= \frac{8}{10} = \frac{4}{5}$.

slope normal $= -5/4$

tangent line

$\Rightarrow y - 4 = \frac{4}{5}(x - 2)$

$5y - 20 = 4x - 8$

$5y = 4x + 12$

Normal line:

$y - 4 = -\frac{5}{4}(x - 2)$

$4y - 16 = -5x + 10$

$4y + 5x = 26$.

$2x^3 - 3y^2 = 8$

$6x^2 - 6y\frac{dy}{dx} = 0$

$x^2 - y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{x^2}{y}$

$\frac{d^2y}{dx^2} = \frac{y(2x) - x^2\frac{dy}{dx}}{y^2}$

$= \frac{2xy - x^2\left(\frac{x^2}{y}\right)}{y^2}$

$= \frac{2xy^2 - x^4}{y^3}$

$x^2 - y\frac{dy}{dx} = 0$

$2x - \left(\frac{dy}{dx}\cdot\frac{dy}{dx} + y\frac{d^2y}{dx^2}\right) = 0$

$2x - \left(\frac{dy}{dx}\right)^2 - y\frac{d^2y}{dx^2} = 0$

$-y\frac{d^2y}{dx^2} = \left(\frac{x^2}{y}\right)^2 - 2x$

$= \frac{x^4 - 2xy^2}{y^2}$

$-\frac{d^2y}{dx^2} = \frac{x^4 - 2xy^2}{y^3}$

$\frac{d^2y}{dx^2} = \frac{2xy^2 - x^4}{y^3}$

Derivatives of Higher Order

Implicit differentiation can also be used to find higher derivative.

Example. *Finding a Second Derivative Implicitly.*

Find d^2y/dx^2 if $2x^3 - 3y^2 = 8$.

Solution. To start, we differentiate both sided of the equation with respect to x in order to find

$$2x^3 - 3y^2 = 8$$
$$6x^2 - 6y\frac{dy}{dx} = 0$$
$$\frac{dy}{dx} = \frac{x^2}{y}, y \neq 0.$$

We can continue with Quotient rule,

$$\frac{d^2y}{dx^2} = \frac{y(2x) - x^2\frac{dy}{dx}}{y^2} = \frac{2xy - x^2\left(\frac{x^2}{y}\right)}{y^2} = \frac{2xy^2 - x^4}{y^3}, y \neq 0.$$

Theorem. *Power Rule for Rational Powers*

If $\frac{p}{q}$ is a rational number, then $x^{\frac{p}{q}}$ is differentiable at every interior point of the domain of $x^{\frac{p}{q}-1}$, and

$$\frac{d}{dx}\left(x^{\frac{p}{q}}\right) = \frac{p}{q}x^{\frac{p}{q}-1}.$$

$\frac{d}{dx}x^n = nx^{n-1}$

$n = p/q$

Example. *Using Rational Power Rule*

(a) $\frac{d}{dx}\left(x^{\frac{1}{2}}\right) = \frac{1}{2}x^{\left(\frac{1}{2}-1\right)} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$, for $x > 0$

$\frac{d}{dx}\left(x^{1/2}\right) = \frac{1}{2x^{1/2}}$

(b) $\frac{d}{dx}\left(x^{\frac{2}{3}}\right) = \frac{2}{3}x^{\left(\frac{2}{3}-1\right)} = \frac{2}{3}x^{-\frac{1}{3}}$, for $x \neq 0$

$\frac{d}{dx}\left(x^{2/3}\right) = \frac{2}{3}x^{-1/3}$

(c) $\frac{d}{dx}\left(x^{-\frac{4}{3}}\right) = -\frac{4}{3}x^{\left(-\frac{4}{3}-1\right)} = -\frac{4}{3}x^{-\frac{7}{3}}$ for $x \neq 0$

$\frac{d}{dx}\left(x^{-4/3}\right) = -\frac{4}{3}x^{-7/3}$

Example. *Using Rational Power and Chain Rule*

(a) $\frac{d}{dx}\left((1-x^2)^{\frac{1}{4}}\right) = \frac{1}{4}(1-x^2)^{-\frac{3}{4}}\cdot(-2x) = -\frac{1}{2}x(1-x^2)^{-\frac{3}{4}}.$

(b) $\frac{d}{dx}(\cos x)^{-\frac{1}{5}} = -\frac{1}{5}(\cos x)^{-\frac{6}{5}}(-\sin x) = \frac{1}{5}\sin x\,(\cos x)^{-\frac{6}{5}}.$